

EXPERIMENT 6

Student Name: Vinay Tiwari

Branch: AIT-CSE AI-ML

Semester: 4

Subject Name: Database Management System

UID: 24BAI70919

Section/Group: 24AIT_KRG-G2

Subject Code: 24CSH-298

Experiment 6 – Understanding and Implementing Cursors for Row-by-Row Data Processing (Oracle, Accenture, Scaler, D. E. Shaw & Co)

Experiment

Experiment 6: Working with PL/SQL cursors to process multiple rows of a table individually. This experiment demonstrates the use of implicit cursors, explicit cursors, and cursor attributes to implement business logic on row-by-row data in Oracle and PostgreSQL.

Aim

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyze how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

Objective

- To implement and analyze implicit cursors for single-row operations.
 - To implement explicit cursors to fetch and process multiple records.
 - To apply cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN) for controlling program flow.
 - To process employee records row by row and apply business logic effectively.
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Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)

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- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

Problem Statement

In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules. Using cursors allows row-by-row processing for detailed business logic implementation.

Practical / Experiment Steps

1. Design PL/SQL programs that demonstrate:
 - The use of implicit cursors for single-row DML operations
 - The use of explicit cursors to fetch and process multiple records
 - The application of cursor attributes to control program execution
2. Create PL/SQL programs to:
 - Fetch employee records from a database table using cursors
 - Process each record individually
 - Display results or apply business logic using cursor attributes

Procedure

1. Open Oracle SQL Developer or pgAdmin and connect to the database.
2. Create an employee table with sample data (id, name, department, salary).
3. Write an anonymous PL/SQL block demonstrating implicit cursor operations.
4. Write a PL/SQL block using an explicit cursor to fetch and process multiple records.
5. Apply cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN to control execution flow.

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6. Execute the blocks and observe the output.
7. Capture results and screenshots for documentation.

Input / Output Details

Input

employee table:

- id (Integer)
- name (Varchar)
- department (Varchar)
- salary (Integer)

Cursor logic used:

- Implicit cursor for single-row processing
- Explicit cursor for multi-row processing
- Cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN)

Step-wise Output

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Step 1 – Create employee Table

```

84 DELETE FROM EMPLOYEE
85 WHERE
86     ID = :ID
87     AND NAME = :NAME
88     AND DEPARTMENT = :DEPARTMENT
89     AND SALARY = :SALARY;
90
91 DESCRIBE EMPLOYEE;
92
93 DELETE FROM EMPLOYEE
94 WHERE
95     ID = :ID
96     AND NAME = :NAME
97     AND DEPARTMENT = :DEPARTMENT
98     AND SALARY = :SALARY;
99
100 DELETE FROM STUDENTS
101 WHERE
102     NAME = :NAME
103     AND MARKS = :MARKS;
104

```

Query result | **Script output** | DBMS output | Explain Plan | SQL history

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.079

```

SQL> DECLARE
-- Declare explicit cursor
CURSOR emp_cursor IS
SELECT id, name, department, salary FROM employee;...
Show more...

```

Step 2 – Insert Data into employee Table

```

4  -- Step 1 - Create employee table
5  CREATE TABLE employee (
6      id NUMBER PRIMARY KEY,
7      name VARCHAR2(50),
8      department VARCHAR2(50),
9      salary NUMBER
10 );

```

Query result | **Script output** | DBMS output | Explain Plan | SQL history

Table EMPLOYEE created.

Elapsed: 00:00:00.014

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Step 3 – Implicit Cursor Operation

```

11
12 -- Step 2 - Insert sample employee data
13 INSERT INTO employee VALUES (1, 'Rahul', 'HR', 25001);
14 INSERT INTO employee VALUES (2, 'Sneha', 'Finance', 32000);
15 INSERT INTO employee VALUES (3, 'Amit', 'IT', 45015);
16 INSERT INTO employee VALUES (4, 'Neha', 'Marketing', 28000);
17 INSERT INTO employee VALUES (5, 'Vikram', 'Sales', 39011);
18
19 COMMIT;

```



Query result **Script output** DBMS output Explain Plan SQL history



SQL> INSERT INTO employee VALUES (2, 'Sneha', 'Finance', 32000)

1 row inserted.

Elapsed: 00:00:00.002

Step 4 – Explicit Cursor Declaration and Fetch

```

23 DECLARE
24     v_salary employee.salary%TYPE;
25 BEGIN
26     -- Implicit cursor: single row query
27     SELECT salary INTO v_salary
28     FROM employee
29     WHERE id = 2;
30
31     DBMS_OUTPUT.PUT_LINE('Salary of employee with ID 2 is: ' || v_salary);
32
33     -- Check if value found (implicit cursor automatically does this)
34     IF SQL%FOUND THEN
35         DBMS_OUTPUT.PUT_LINE('Record found.');
```



Query result **Script output** DBMS output Explain Plan SQL history



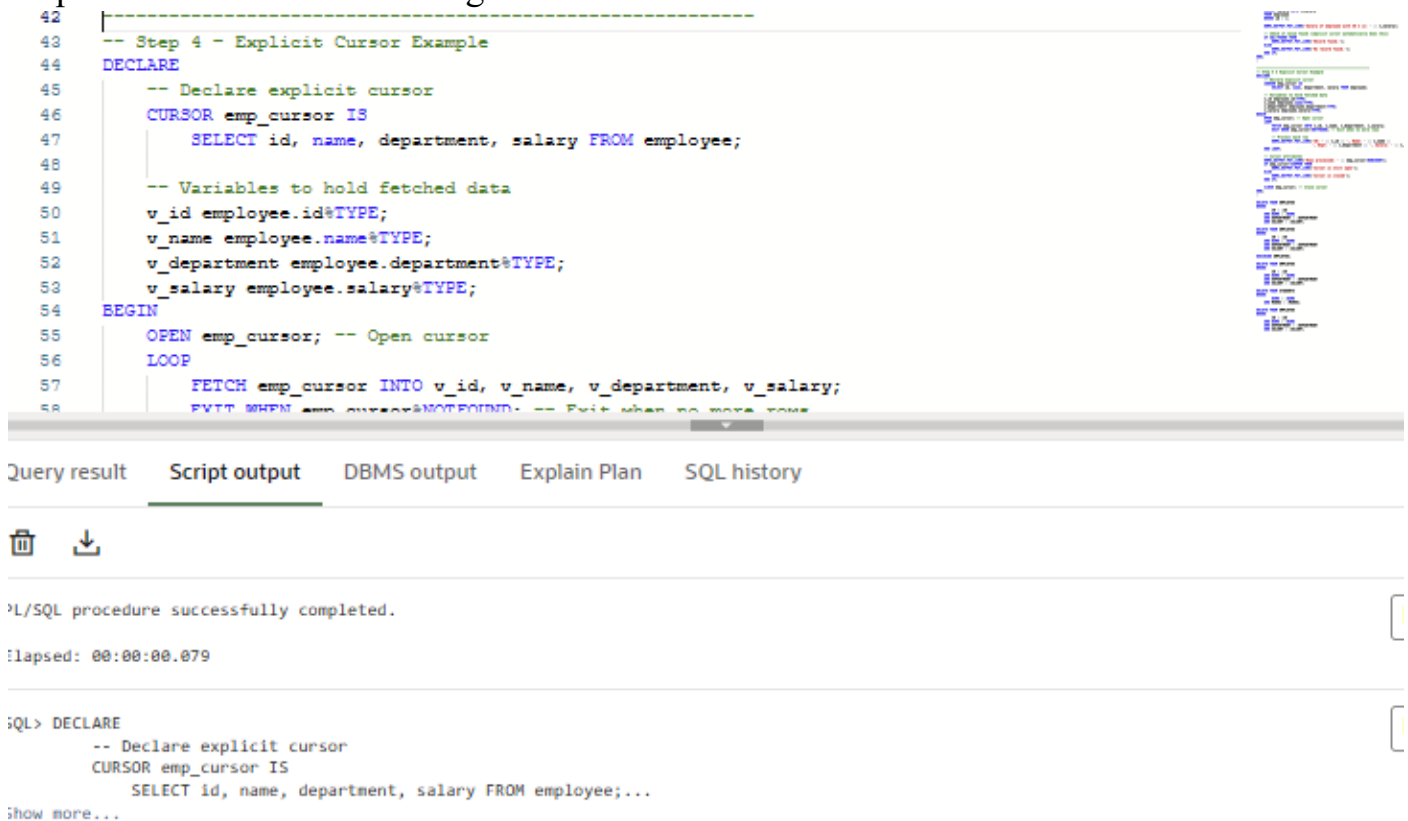
SQL> DECLARE
v_salary employee.salary%TYPE;
BEGIN
-- Implicit cursor: single row query...

Salary of employee with ID 2 is: 32000
Record found.



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Step 5 – Process Records Using Cursor Attributes



```

42 |-----
43 |-- Step 4 - Explicit Cursor Example
44 |DECLARE
45 |    -- Declare explicit cursor
46 |    CURSOR emp_cursor IS
47 |        SELECT id, name, department, salary FROM employee;
48 |
49 |    -- Variables to hold fetched data
50 |    v_id employee.id%TYPE;
51 |    v_name employee.name%TYPE;
52 |    v_department employee.department%TYPE;
53 |    v_salary employee.salary%TYPE;
54 |BEGIN
55 |    OPEN emp_cursor; -- Open cursor
56 |    LOOP
57 |        FETCH emp_cursor INTO v_id, v_name, v_department, v_salary;
58 |        EXIT WHEN emp_cursor%NOTFOUND; -- Exit when no more rows
  
```

Query result Script output DBMS output Explain Plan SQL history

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.079

SQL> DECLARE
 -- Declare explicit cursor
 CURSOR emp_cursor IS
 SELECT id, name, department, salary FROM employee;...
 show more...

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the role of cursors in PL/SQL for handling multi-row query results.
- Differentiate between implicit cursors and explicit cursors.
- Use cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN.
- Develop PL/SQL programs that process database records row by row.
- Apply cursor-based logic in real-world business scenarios used in organizations like Oracle, Accenture, Scaler, and D. E. Shaw & Co.